

his famous 1919 paper '*A Method of Reaching Extreme Altitudes*'. Much of the basis for modern space technology still comes from this paper.

Meanwhile, German physicist Hermann Oberth was also studying how to send rockets into space and for his doctoral thesis, wrote *Die Rakete zu den Planetenräumen* (*By Rocket into Planetary Space*). The thesis was rejected, but Oberth self-published it as a book that found great admiration and following throughout Germany as well as Russia.

For the next 20 years, scientists from the US, Germany and Russia were at the forefront of research on rockets and space travel – a competition that was only enhanced once World War II started.

The V-2 rocket

One would think that with all the attention that the Space Race received during the Cold War, either the US or Russia would have been the pioneer of space exploration. But surprisingly, the Germans were at the forefront.

In the 1930s and the 1940s, one name became synonymous with space research across the world: Werner Von Braun. Von Braun was greatly influenced by the works of both, Goddard and Oberth, and started to work on liquid-fuelled rockets from an early age. In the 1930s, when the Nazi party came into power, Von Braun was enlisted by the army and by 1934, he had successfully launched two liquid-fuelled rockets, although neither of them reached space. This series of rockets would be known as the Aggregat series, and titled A-1, A-2 and so on.

Of course, Hitler's interest in the rockets was to use them as long-range ballistic missiles that would have a reach unparalleled by any other artillery on the planet. He personally met with Von Braun a few times and finally gave the go-ahead for A-4.

It was this fourth Aggregat rocket that would turn out to be the most significant in mankind's history. More famously known as the V-2 rocket, this would be the first man-made machine to reach 100 kilometres from the surface of the earth – the demarcating line between Earth's atmosphere and outer space.

There were some key elements and advancements in the V-2 rocket which no American or Russian scientists had managed at the time. The liquid-fuel, of course, was the most important part. The rocket used an ethanol-water mixture as fuel and liquid oxygen as the oxidizer, a combination of which allowed it to reach heights of 80 kilometres – high enough to be able to attack London.

More importantly, being a combat ballistic missile, the rocket needed